PYL7167 / AMP 7167

Atomic Physics — Quiz Study Notes

Readable PDF: formulae, theory, and all 18 practice solutions

From your lecture notes, the practice set (high emphasis), and Haken & Wolf, Bransden & Joachain, White, Foot, Kittel, Jackson.

How to use this file tonight.

1. Memorise Part I (formula sheet), 30 minutes.
2. Skim Part II only on topics you are weak on.
3. Work Part III on paper first, then check. Do not skip 9, 11, 12, 15, 17, 18.
4. In the morning, read only Part IV.

If you have two hours: Part I, problems 1–8 and 18, then diagrams in 9, 12, 16, then Part IV.

Contents

1 Formula sheet

Constants to memorise

Use these unless the paper gives others

$$R_\infty = 109\,737.318\ \text{cm}^{-1}, \qquad R_H \approx 109\,677.6\ \text{cm}^{-1},$$
$$hcR_\infty = 13.6\ \text{eV}, \qquad a_0 = 0.529\ \text{Å},$$
$$\alpha = 1/137, \qquad c = 137\ \text{a.u.},$$
$$\mu_B = e\hbar/2m_e, \qquad m_p/m_e \approx 1836,$$
$$M_D/m_e \approx 3670, \qquad k_B = 8.617\times10^{-5}\ \text{eV}\,\text{K}^{-1}.$$

Sodium D lines: D_2: $3\,^2P_{3/2} \to 3\,^2S_{1/2}$ at **589.0 nm**; D_1: $3\,^2P_{1/2} \to 3\,^2S_{1/2}$ at **589.6 nm**.

1.1 Hydrogenic energies and spectra

Bohr / Schrödinger energy (reduced mass μ):

$$E_n = -\frac{\mu Z^2 e^4}{2(4\pi\epsilon_0)^2\hbar^2}\frac{1}{n^2} = -13.6\ \text{eV}\cdot\frac{\mu}{m_e}\cdot\frac{Z^2}{n^2}.$$

Rydberg formula:

$$\frac{1}{\lambda} = R_M Z^2\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right), \qquad R_M = R_\infty\frac{\mu}{m_e} = \frac{R_\infty}{1+m_e/M},$$

$$R_\infty = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} = 109\,737.318\ \text{cm}^{-1}.$$

Series limit ($n_i\to\infty$): $1/\lambda_\infty = R_M Z^2/n_f^2$. Longest wavelength in a series: $n_i = n_f+1$. Shortest = series limit.

Series	n_f	λ range (H)	Region	Overlap?
Lyman	1	91.2–121.6 nm	UV	no overlap with Balmer
Balmer	2	364.6–656.3 nm	vis $\to$ near UV	no overlap with Paschen
Paschen	3	820.4–1875 nm	IR	**overlaps Brackett**
Brackett	4	1458–4051 nm	IR	**overlaps Paschen and Pfund**
Pfund	5	2279–7460 nm	IR	**overlaps Brackett**

Correspondence principle ($n \gg 1$, $\Delta n = p$ small):

$$\frac{1}{\lambda} \approx (\text{const})\frac{p}{n^3}, \qquad \omega_{\text{orb}} = 2\pi\cdot 2Rc/n^3.$$

Equating classical $E(\omega)$ to $E_n = -Rch/n^2$ recovers R_∞.

Exotic atoms / excitons — same formulae with the actual reduced mass and dielectric constant:

$$\mu_{\pi p} = \frac{m_\pi m_p}{m_\pi + m_p}, \qquad R_{\text{ex}} = \frac{13.6\ \text{eV}}{\varepsilon_r^2}\frac{\mu}{m_e}.$$

Wannier exciton: $E_n = E_g - R_{\text{ex}}/n^2$. Absorption: hydrogen-like discrete lines plus continuum for $h\nu > E_g$.

1.2 Atomic units, sizes, virial

$$a_0 = 0.529\ \text{Å}, \quad 1\ \text{Hartree} = 27.2\ \text{eV} = 2\times 13.6\ \text{eV}, \quad v_1 = \alpha c = c/137.$$

$$\langle r\rangle_{nlm} = \frac{a_0 n^2}{Z}\left\{1+\frac{1}{2}\left[1-\frac{\ell(\ell+1)}{n^2}\right]\right\}, \qquad \left\langle\frac{1}{r}\right\rangle_{nlm} = \frac{Z}{a_\mu n^2}.$$

Most probable radius $\neq \langle r\rangle$ (the radial tail pulls $\langle r\rangle$ out). For 1s: $r_{\rm mp} = a_0/Z$, $\langle r\rangle = 1.5\,a_0/Z$. Rydberg atom: $r_n = n^2 a_0$, $E_b = 13.6\ \text{eV}/n^2$, **weakly** bound.

Virial for $V = \alpha r^k$: $2\langle T\rangle = k\langle V\rangle$. Coulomb $k=-1$: $\langle T\rangle = -E$, $\langle V\rangle = 2E$. Harmonic oscillator $k=2$: $\langle T\rangle = \langle V\rangle$.

Small-r radial behaviour: $R_{nl}(r) \propto r^\ell$ (the function $u = rR \propto r^{\ell+1}$).

Unsold: $\sum_{m=-\ell}^{\ell} |Y_{\ell m}|^2 = (2\ell+1)/4\pi$ (angle-independent). Closed shells are spherically symmetric.

1.3 Alkali atoms and quantum defect

The central field is still spherical (so m remains degenerate) but **not** pure Coulomb, so ℓ-degeneracy dies.

$$E_{n\ell} = -R\,hc/(n-\delta_{n\ell})^2.$$

Penetration: $\delta_s \gg \delta_p \gg \delta_d \approx \delta_f \approx 0$. For Li (your notes): $\delta_s \approx 0.4$, $\delta_p \approx 0.04$, $\delta_{d,f} \approx 0$. Ground state of Li: $1s^2 2s\ {}^2S_{1/2}$. Resonance line $2p \to 2s$. Why $\ell < n$: radial quantum number $n_r = n - \ell - 1 \geq 0$.

1.4 Angular momentum, terms, coupling

$$\mathbf{J} = \mathbf{L} + \mathbf{S}, \qquad \mathbf{L}\cdot\mathbf{S} = \frac{1}{2}(J^2 - L^2 - S^2).$$

Term symbol: ${}^{2S+1}L_J$ with $L = S, P, D, F, \ldots$ ($L = 0, 1, 2, 3, \ldots$).

LS (light atoms): $\mathbf{L} = \sum \mathbf{l}_i$, $\mathbf{S} = \sum \mathbf{s}_i$, then $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

jj (heavy atoms): $\mathbf{j}_i = \mathbf{l}_i + \mathbf{s}_i$, then $\mathbf{J} = \sum \mathbf{j}_i$.

Two nonequivalent electrons $\ell_1 = 1$, $\ell_2 = 2$: $L = 1,2,3$, $S = 0,1 \Rightarrow {}^1P, {}^1D, {}^1F, {}^3P, {}^3D, {}^3F$, with $J = |L-S| \ldots L+S$.

Hund (LS, equivalent electrons): (1) max S; (2) then max L; (3) $J_{\min}$ if shell < half full, $J_{\max}$ if > half full.

jj equivalent electrons $j = 5/2$:

$$(5/2)^{1,5}: \ J = 5/2; \quad (5/2)^{2,4}: \ J = 0,2,4; \quad (5/2)^3: \ J = 3/2, 5/2, 9/2; \quad (5/2)^6: \ J = 0.$$

Hole–particle: $(j)^{2j+1-N}$ has the same J list as $(j)^N$.

Magnetic moment (weak B):

$$\boldsymbol{\mu}_J = -g_J \mu_B \mathbf{J}/\hbar, \qquad g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)},$$

$$\mu = g_J\sqrt{J(J+1)}\,\mu_B, \qquad \Delta E = g_J \mu_B B m_J.$$

$g_L = 1$, $g_S = 2.0023 \approx 2$. Gyromagnetic: $\omega_L = \gamma B$, $\gamma_L = e/2m$, $\gamma_S \approx e/m$.

Helium: para $S=0$ (singlets, including GS $1s^2\ {}^1S_0$); ortho $S=1$ (triplets). Intercombination lines are E1-forbidden $\Rightarrow$ two superimposed spectra.

1.5 Fine structure (spin–orbit)

Internal **B** of orbital motion + Thomas factor 1/2:

$$V_{LS} = \frac{Ze^2\mu_0}{8\pi m_e^2 r^3}\,\mathbf{S}\cdot\mathbf{L} = \frac{a}{2}[j(j+1) - \ell(\ell+1) - s(s+1)],$$

$$\left\langle \frac{1}{r^3} \right\rangle = \frac{Z^3}{a_H^3 n^3 \ell(\ell+\frac{1}{2})(\ell+1)} \qquad (\ell \geq 1).$$

Interval between $j = \ell + \frac{1}{2}$ and $j = \ell - \frac{1}{2}$:

$$\Delta E_{\rm fs} \propto \frac{Z^4}{n^3\,\ell(\ell+1)}.$$

Dirac / rel. QM: $E = E(n, j)$ not $E(n, \ell)$. Lamb shift (QED) splits $2S_{1/2}$ **above** $2P_{1/2}$ by 1057.8 MHz $\approx$ 0.035 cm^{-1} (this is **not** 0.031 eV). $2S_{1/2}$ is metastable (E1 $2S \to 1S$ forbidden, $\Delta\ell = 0$).

1.6 External fields

Stern–Gerlach.

$$U = -\boldsymbol{\mu} \cdot \mathbf{B}, \qquad F_z \approx \mu_z \frac{\partial B_z}{\partial z}.$$

$^2S_{1/2}$ alkali: 2 beams, $\mu_z = \pm\mu_B$. Cannot be done with free electrons (Lorentz force swamps the spin force). Deflection of one component after magnet length L plus drift ℓ, speed v:

$$s = \frac{\mu_z}{M} \frac{\partial B_z}{\partial z} \frac{L}{v^2} \left(\ell + \frac{L}{2}\right), \qquad \text{spot separation } = 2s.$$

Use $v = \sqrt{2kT/M}$ unless told otherwise.

Zeeman Hamiltonian.

$$H' = \frac{\mu_B}{\hbar}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B} = \mu_B B(L_z + 2S_z)/\hbar.$$

Regime	Criterion	Energies	Pattern
Weak / anomalous	$\mu_B B \ll E_{\text{fs}}$	$g_J \mu_B B m_J$	many uneven components
Normal	$S = 0$ so $g = 1$	$\mu_B B m_L$	Lorentz triplet
Paschen–Back	$\mu_B B \gg E_{\text{fs}}$	$\mu_B B(m_L + 2m_S)$	normal triplet restored

E1 selection: $\Delta m_J = 0$ (π, linear $\parallel$ $\mathbf{B}$), $\Delta m_J = \pm 1$ (σ, circular about $\mathbf{B}$). Also $\Delta J = 0, \pm 1$ (not $0 \to 0$), $\Delta L = \pm 1$, $\Delta S = 0$, and $\Delta m_S = 0$ in Paschen–Back.

Landé g values you will need:

Term	g_J
$^2S_{1/2}$	2
$^2P_{1/2}$	2/3
$^2P_{3/2}$	4/3
$^2D_{3/2}$	4/5
$^2D_{5/2}$	6/5
$^1P_1, {}^1D_2$ (singlets)	1

Linear Stark (hydrogen only, first order) because of ℓ-degeneracy:

$$\Delta E = \frac{3}{2} n(n_1 - n_2)\, e a_0\, \mathcal{E}, \qquad n = n_1 + n_2 + |m| + 1.$$

$n = 2$: three levels, $\Delta E = 0, \pm 3\, e a_0 \mathcal{E}$. $n = 3$: five levels, $\Delta E = 0, \pm\frac{9}{2} e a_0 \mathcal{E}, \pm 9\, e a_0 \mathcal{E}$. Alkali / non-H: only **quadratic** Stark. Polarisation: π if $\Delta m = 0$, σ if $\Delta m = \pm 1$.

1.7 Radiation, Einstein, selection rules

Semiclassical: field classical, atom quantum. Coulomb gauge $\nabla \cdot \mathbf{A} = 0$.

$$H = \frac{1}{2m}(\mathbf{p} + e\mathbf{A})^2 + V.$$

Drop A^2 at low intensity (keeping it gives multiphoton processes). Dipole (E1): $e^{i\mathbf{k}\cdot\mathbf{r}} \approx 1$, rate $\propto |\hat{\epsilon} \cdot \mathbf{r}_{ba}|^2$.

E1 selection:

$$\Delta\ell = \pm 1, \quad \Delta m = 0, \pm 1, \quad \Delta S = 0, \quad \Delta J = 0, \pm 1 \ (J = 0 \not\to J = 0), \quad \text{parity changes.}$$

$\Delta m = 0$: linear (z, π). $\Delta m = \pm 1$: circular ($\sigma_\pm$, $x \pm iy$). Dipole-forbidden $\neq$ strictly forbidden (M1, E2, two-photon, collisions).

Einstein (energy density per Hz):

$$A_{21} = \frac{8\pi h\nu^3}{c^3} B_{21}, \qquad B_{12} = \frac{g_2}{g_1} B_{21}.$$

Stimulated rate $\propto$ intensity. Spontaneous A is not.

Natural (Lorentzian) FWHM of a line to a stable lower level:

$$\Delta E = \hbar/\tau, \qquad \Delta\nu = \frac{1}{2\pi\tau}, \qquad \Delta\lambda = \frac{\lambda^2}{c}\Delta\nu.$$

$\tau =$ **mean** life. If half-life is given, $\tau = t_{1/2}/\ln 2$.

1.8 Line shapes

Doppler (Gaussian) FWHM:

$$\Delta\nu_D = \frac{2\nu_0}{c}\sqrt{\frac{2kT\ln 2}{M}}, \qquad \Delta\lambda_D = \frac{2\lambda_0}{c}\sqrt{\frac{2kT\ln 2}{M}},$$

$$I(\nu) = I(\nu_0)\exp\left[-\frac{Mc^2}{2kT}\left(\frac{\nu-\nu_0}{\nu_0}\right)^2\right].$$

Doppler **shift** of one velocity class: $\Delta\nu/\nu = v/c$. Doppler **width** is the FWHM of the thermal envelope. Pressure broadening: collisions, can be asymmetric, and shortens lifetime.

Lecture side-notes

- Optical electron = valence electron that is excited. Equivalent electrons share (n, ℓ).
- Line spectra: atoms. Band: molecules/solids. Continuous: black-body / free–bound.
- Photocell: reverse-biased pn, $h\nu > E_g$. Solar cell: same physics, **no** bias.
- Auger kinetic energy is a difference of binding energies; independent of the photon that dug the hole.
- King: $|q_p + q_e| \lesssim 10^{-20} e$.
- Classical radiative collapse of H: Larmor, $T \sim 10^{-11}$ s (not ms).
- Relativistic mass correction to R is not enough for $Z \gtrsim 20$.

2 Theory notes

2.1 What an "atom" means in this course

One-electron (hydrogenic) atoms: a nucleus of charge $+Ze$ plus exactly one electron. Examples: H, He^+, Li^{2+}, C^{5+}. Energies scale as Z^2, lengths as $1/Z$.

Optical (valence) electrons: the electrons that actually change state in optical spectra. Na has one; Ca has two.

Equivalent electrons: same n and ℓ. Helium $1s^2$ is two equivalent s electrons.

Three kinds of spectra: **line** (atoms), **band** (molecules/solids), **continuous** (free–bound or black-body).

Helium looks like two superimposed spectra: *parahelium* (singlets, $S = 0$, includes the ground state $1s^2\,{}^1S_0$) and *orthohelium* (triplets, $S = 1$, lowest term $1s2s\,{}^3S_1$, metastable). E1 cannot change S.

2.2 Bohr model, Rydberg constant, correspondence principle

Classical circular orbit: Coulomb = centripetal

$$\frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2} = mr\omega^2 \quad\Rightarrow\quad E = T + V = -\frac{1}{2}\cdot\frac{e^2}{4\pi\epsilon_0 r}.$$

Eliminating r in favour of ω:

$$E = -\frac{1}{2}(4\pi\epsilon_0)^{-2/3}(e^4 m\omega^2)^{1/3}.$$

Quantum frequencies of neighbouring levels at large n:

$$\nu = Rc\left(\frac{1}{n^2} - \frac{1}{(n+1)^2}\right) \approx 2Rc/n^3, \qquad \omega = 2\pi\nu.$$

Match $E_n = -Rch/n^2$ to the classical $E(\omega)$ and you recover $R_\infty = m_e e^4/(8\varepsilon_0^2 h^3 c)$.

Experiment: $R_H = 109\,677.58\ \mathrm{cm}^{-1}$. Infinite-mass theory: $R_\infty = 109\,737.32\ \mathrm{cm}^{-1}$. The gap $\approx 60\ \mathrm{cm}^{-1}$ is **reduced mass**:

$$R_H = \frac{R_\infty}{1 + m_e/M_p}.$$

This fix is not enough for $Z \gtrsim 20$: fine structure and QED grow as Z^4 and higher.

Resonance line of H: Lyman-α, $n = 2 \to 1$, 121.6 nm.

Classical collapse (Larmor): an accelerating charge radiates $P = \mu_0 e^2 a^2/(6\pi c)$. Time to spiral from a_0 into the nucleus is $\sim 10^{-11}$ s — not milliseconds. Bohr stops this by forbidding continuous radiation in a stationary state.

2.3 Schrödinger hydrogen: degeneracy, radial functions, virial

Central potential $\Rightarrow [H, \mathbf{L}] = [H, L^2] = 0$, so $\{H, L^2, L_z\}$ share eigenfunctions

$$\psi_{nlm}(\mathbf{r}) = R_{nl}(r)\,Y_{\ell m}(\theta, \phi).$$

- Energy of a pure Coulomb problem depends only on n (**accidental / ℓ-degeneracy**), degeneracy n^2 without spin, $2n^2$ with spin.
- m-degeneracy is from **spherical symmetry** (any central $V(r)$). It survives in alkalis; ℓ-degeneracy does not.

Why $\ell < n$: bound Coulomb solutions exist only for $n_r = 0, 1, 2, \ldots$ with $n = n_r + \ell + 1$.

Near the origin: $R_{nl}(r) \to r^\ell$. ($u = rR \propto r^{\ell+1}$. If a T/F says "$R_{nl} \propto r^{\ell+1}$", it is false.)

Useful functions:

$$R_{10}(r) = 2\left(\frac{Z}{a_0}\right)^{3/2} e^{-Zr/a_0}, \qquad R_{21}(r) = \frac{1}{\sqrt{3}}\left(\frac{Z}{2a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0}.$$

Larger Z pulls $|R|^2$ inward. s waves have $R(0) \neq 0$ (electron capture possible).

Radial probability $P(r) = r^2|R_{nl}|^2$. Maxima from $dP/dr = 0$, not from $\langle r \rangle$. $\langle r \rangle > r_{\mathrm{mp}}$ because $P(r)$ has a long outer tail.

Virial: Coulomb $\langle T \rangle = +13.6$ eV, $\langle V \rangle = -27.2$ eV, $E = -13.6$ eV for H 1s.

Unsold's theorem: a closed subshell is spherically symmetric. Proof for $\ell = 1$ is problem 17.

2.4 Alkali atoms, quantum defect, ionisation

An alkali is hydrogen-like **outside** a closed noble-gas core. The valence electron sees $V \sim -e^2/4\pi\epsilon_0 r$ far away (core charge +1), and $V \sim -Ze^2/4\pi\epsilon_0 r$ if it penetrates the core.

> Do **not** write "the potential is not spherically symmetric". That would split m, which it does not. Spherical symmetry is intact; Coulomb degeneracy is not.

Quantum defect $\delta_{n\ell}$ absorbs core penetration and polarisation:

$$E_{n\ell} = -R_{\mathrm{atom}}hc/(n - \delta_{n\ell})^2, \qquad n^* = n - \delta.$$

δ is nearly independent of n for a given ℓ, and falls rapidly with ℓ.

Term diagram vs hydrogen: every ns level of Li sits **below** the H level of the same n; np slightly below; nd, nf almost on top of H. Ground state is $2s$, not $1s$. Allowed arrows: $\Delta\ell = \pm 1$.

Ionisation energies jump when you start breaking a closed shell (Li: 5.4 eV then ~ 75 eV).

2.5 Fine structure, relativistic QM, Lamb shift

In the electron's rest frame the nucleus orbits, producing $\mathbf{B}_L \propto \mathbf{L}/r^3$. Then $V = -\boldsymbol{\mu}_s \cdot \mathbf{B}_L$ with a Thomas factor 1/2. On $|n\ell j m_j\rangle$,

$$\mathbf{L} \cdot \mathbf{S} = \frac{\hbar^2}{2}\left[j(j+1) - \ell(\ell+1) - s(s+1)\right],$$

so each Bohr level with $\ell \geq 1$ splits into $j = \ell \pm \frac{1}{2}$. The doublet interval scales as $Z^4/[n^3\ell(\ell+1)]$.

Hydrogen $2p$ interval is 0.365 cm^{-1} ($\sim 4.5 \times 10^{-5}$ eV). A problem that quotes "0.3 eV" is a **scaling exercise**, not a measured number.

Dirac hydrogen: $E = E(n, j)$. Thus $2S_{1/2}$ and $2P_{1/2}$ are still degenerate. Lamb–Retherford (1947): microwave resonance on a hot H beam; $2S_{1/2}$ is metastable and survives a 10 cm flight; extrapolate to $B = 0$. Result: $2S_{1/2}$ lies 1057.8 MHz **above** $2P_{1/2}$. This is a QED radiative correction, not predicted by Dirac. Frequency 1057 MHz ≈ 4.4 μeV ≈ 0.035 cm^{-1}. If lecture notes say "0.031 eV", that is a unit slip.

Metastable $2S_{1/2}$ of H: E1 needs $\Delta\ell = \pm 1$; there is no lower $\ell = 1$ state below $2S$. Decay is two-photon, lifetime ~ 0.12 s. ($2P$ lives 1.6 ns.)

2.6 Magnetic moments, Larmor, Stern–Gerlach

Orbital: $\boldsymbol{\mu}_L = -(e/2m_e)\mathbf{L}$, $g_L = 1$. Spin: $\boldsymbol{\mu}_S = -g_S\mu_B\mathbf{S}/\hbar$, $g_S \approx 2$. Because $g_S \neq g_L$, $\boldsymbol{\mu}_J$ is not simply antiparallel to $\mathbf{J}$; the surviving projection is the Landé one.

Torque $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B} \Rightarrow$ Larmor precession $\omega_L = \gamma B$, independent of the angle α between $\boldsymbol{\mu}$ and $\mathbf{B}$.

SG force in an inhomogeneous field: $\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B})$. With the standard pole-piece geometry, $F_z = \mu_z \partial B_z/\partial z$. Alkali GS $^2S_{1/2}$: two spots. A $J = 3/2$ beam would give four spots. Free electrons fail because of $e\mathbf{v} \times \mathbf{B}$. Use neutral atoms (Ag, Cu, H, Na, K, Cs).

2.7 Zeeman, Paschen–Back, and Stark

Full magnetic Hamiltonian (one electron, ignoring diamagnetism):

$$H = H_0 + \zeta\, \mathbf{L} \cdot \mathbf{S} + \frac{\mu_B}{\hbar}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B}.$$

"Weak" vs "strong" is relative to the internal FS interval of that atom.

Normal Zeeman ($S = 0$, e.g. Cd $^1D_2 \to {}^1P_1$): $g = 1$ on both levels. Selection $\Delta m = 0, \pm 1$ yields the Lorentz triplet.

Anomalous Zeeman ($S \neq 0$): each FS level fans into $2J+1$ sublevels with spacing $g_J \mu_B B$. Because $g_{\text{upper}} \neq g_{\text{lower}}$, you get many components with unequal spacing. Sodium D: $D_2 = {}^2P_{3/2} \to {}^2S_{1/2}$ (stronger); $D_1 = {}^2P_{1/2} \to {}^2S_{1/2}$.

Paschen–Back: **L** and **S** decouple. Good quantum numbers m_L, m_S. Energies $E = \mu_B B(m_L + 2m_S)$. Optical selection $\Delta m_L = 0, \pm 1$, $\Delta m_S = 0$ collapses the forest back to a normal triplet.

Stark: $H' = e\mathcal{E}z$. Hydrogen has a **linear** (first-order) Stark effect because of ℓ-degeneracy. Non-hydrogenic atoms have only quadratic Stark ($\langle z \rangle = 0$ by parity). Polarisation: π ($\Delta m = 0$) and σ ($\Delta m = \pm 1$).

2.8 Interaction with light

Treat $\mathbf{E}, \mathbf{B}$ classically and the atom quantum-mechanically.

$$i\hbar \partial_t \Psi = \left[\frac{1}{2m} (-i\hbar \nabla + e\mathbf{A})^2 + V(r) \right] \Psi.$$

Identify $H_0 = p^2/2m + V$ and $H' \approx (e/m)\mathbf{A} \cdot \mathbf{p}$. First-order TDPT, long-time limit, produces a rate $\propto$ (intensity) $\times |M_{ba}|^2$.

Dipole approximation: $\lambda \gg a_0$, so $e^{i\mathbf{k}\cdot\mathbf{r}} \approx 1$ and $M \propto \hat{\epsilon} \cdot \mathbf{r}_{ba}$. Angular integrals of $z = r\cos\theta$ force $\Delta m = 0$; $x \pm iy$ force $\Delta m = \pm 1$. Parity of $Y_{\ell m}$ is $(-1)^\ell$, so E1 changes parity.

Einstein A (spontaneous) is not in semiclassical Maxwell theory; QM derives B, then $A_{21} = (8\pi h \nu^3 / c^3) B_{21}$. Absorption and stimulated emission share the same $|M|^2$. Scattering is a second-order two-photon process; absorption is first-order.

2.9 Widths and shapes

An excited state with mean life τ is not monochromatic: $\Delta E \, \Delta t \sim \hbar$ with $\Delta t \sim \tau$.

Mechanism	Shape	Scales as	Notes
Natural	Lorentzian	$1/\tau$	homogeneous; independent of T
Doppler	Gaussian	$\sqrt{T/M}\,\nu_0$	inhomogeneous; thermal v_x
Pressure / collision	often Lorentzian, can be asymmetric	density $\times \sigma \times v$	shortens effective τ

Rule of thumb: optical Doppler at 300 K is GHz; natural is MHz (Na D_2: $\tau = 16$ ns $\Rightarrow \Delta\nu \approx 10$ MHz).

2.10 Multi-electron bookkeeping

LS coupling (light atoms, residual Coulomb $\gg$ spin–orbit): Pauli-allowed (S, L) terms, then $J = |L - S| \ldots L + S$, then Hund.

jj coupling (heavy atoms): each electron has a good $j = \ell \pm 1/2$; equivalent-j occupations restricted by Pauli on m_j. Use hole–particle symmetry.

Platinum [Xe]$4f^{14}5d^9 6s^1$: a d-hole plus an s electron $\Rightarrow {}^3D$ and ${}^1D$. More-than-half $d^9 \Rightarrow$ inverted FS, GS ${}^3D_3$. Then $g = 4/3$, $\mu = (8\sqrt{3}/3)\mu_B$, weak-field span $= 8\mu_B B$.

Helium spin: the triplet $S = 1$ subspace is $\{|\uparrow\uparrow\rangle, \ (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}, \ |\downarrow\downarrow\rangle\}$. Problem 15 is the $|\downarrow\downarrow\rangle$ case done with Pauli matrices.

2.11 Lecture odds and ends

Photocell vs solar cell. Both: $h\nu > E_g$ creates e–h pairs at a pn junction. Photocell is reverse biased. Solar cell has no bias.

Faraday cup. Measures beam current. A suppressor electrode throws secondary electrons back into the cup.

Auger / Coster–Kronig. A core hole is filled by an electron from a higher shell; the energy released ejects another electron. Kinetic energy $= E_{\text{hole}} - E_1 - E_2$, independent of the photon that made the hole. Hydrogen has only one electron, so no Auger.

Photoelectric trap. Threshold is a **frequency** condition $h\nu > W_0$, not an energy-density condition. A very intense red beam still emits nothing if $\nu < \nu_0$.

Polarisation of Zeeman components. Viewed along **B**: only $\sigma_\pm$, circular. Viewed perpendicular: π linear $\parallel$ **B**, σ linear $\perp$ **B**.

2.12 Exotic hydrogenic systems

Anything bound by Coulomb physics is hydrogen with a new μ and perhaps a dielectric ε_r:

$$E_n = -13.6 \text{ eV} \cdot \frac{\mu}{m_e} \cdot \frac{Z^2}{n^2 \varepsilon_r^2}, \qquad a = \frac{\varepsilon_r m_e}{\mu Z} a_0.$$

Pionic atom $p^+\pi^-$: $\mu = m_\pi m_p/(m_\pi + m_p)$. With $m_\pi = 250\, m_e$ you **must** reduce against the proton; $\mu \approx 220\, m_e$, not $250\, m_e$. Rydberg atom: $n \gg 1$, radius $n^2 a_0$, binding $13.6/n^2$ eV, neighbouring spacing $\approx 27.2/n^3$ eV. Weakly bound. Wannier exciton: e–h pair in a semiconductor, $\varepsilon_r \sim 10$, $\mu \sim 0.1$–$0.7\, m_e$. Rydberg collapses from eV to meV.

3 Practice problems, fully worked

Cover the answer, solve on paper, then check. Constants as in Part I.

One-line key

1. $6856.71\ \text{cm}^{-1} = 1458.4$ nm (use R_D, not R_∞)
2. 104 eV (reduced mass $220\, m_e$)
3. $\Delta\nu = 9.95$ MHz, $\Delta E = 4.11 \times 10^{-8}$ eV
4. Lyman/Balmer: no. Paschen/Brackett/Pfund: yes (IR)
5. $\lambda_2 \approx 600, 595, 593, 592$ nm; $\delta E_1 + \delta E_2 \lesssim 1.1 \times 10^{-4}$ eV
6. $R_{\text{ex}} = 95.2$ meV; $n = 2..5$: 23.8, 10.6, 5.95, 3.81 meV
7. 1.555 eV (Z^4/n^3)
8. H: 2.04 Å; D: 1.45 Å
9. 3 FS lines; $g = 6/5, 4/5, 4/3, 2/3$; PB $\rightarrow$ Lorentz triplet
10. ${}^{1,3}P, D, F$
11. $P_2P_3 \approx 10$ cm
12. $n = 3$: 5 levels at $0, \pm 9/2, \pm 9$ ($ea_0\mathcal{E}$)
13. GS ${}^3D_3$, $\mu = (8\sqrt{3}/3)\mu_B$, span $8\mu_B B$
14. $J = 0, 2, 4$
15. $S_z\chi = -\hbar\chi$, $S^2\chi = 2\hbar^2\chi$
16. ns pulled down most; arrows $2p \rightarrow 2s$ and $3d \rightarrow 2p$
17. sum $= 3/4\pi$
18. All seven false

3.1 Problem 1 — Series limit of deuterium, $n = 4$

This is the Brackett series. The series limit is the ionisation threshold from $n = 4$:

$$\bar{\nu}_\infty = \frac{1}{\lambda_\infty} = R_D \cdot \frac{1}{4^2} = \frac{R_D}{16}.$$

You **must** convert $R_\infty \rightarrow R_D$. The notes spent a page on this.

$$R_D = \frac{R_\infty}{1 + m_e/M_D}, \qquad \frac{M_D}{m_e} \approx 3670.48 \quad \text{(deuteron, not } 2m_p\text{)}.$$

$$R_D = 109\,737.318 \times \frac{3670.48}{3671.48} = 109\,707.43\ \text{cm}^{-1}.$$

$$\bar{\nu}_\infty = \frac{109\,707.43}{16} = 6856.71\ \text{cm}^{-1}, \qquad \lambda_\infty = 1458.4\ \text{nm} = 14\,584\ \text{Å}.$$

Write both wavenumber and wavelength. Trap: using R_∞ raw gives $6858.58\ \text{cm}^{-1}$. Using R_H (protium) is also wrong.

3.2 Problem 2 — Pionic atom, $n = 4$ vs $n = 6$

Hydrogenic with reduced mass

$$\mu = \frac{m_\pi m_p}{m_\pi + m_p} = \frac{250 \times 1836}{250 + 1836} m_e = 220.0\, m_e.$$

The proton is *not* infinitely heavy compared with the pion. The parenthetical "use $m_\pi = 250\, m_e$" gives you the pion mass, not permission to skip μ.

$$\Delta E = E_6 - E_4 = 13.6 \times 220.0 \times \left(\frac{1}{16} - \frac{1}{36}\right) = 13.6 \times 220.0 \times \frac{5}{144} = 103.9 \text{ eV} \approx 104 \text{ eV}.$$

Naive replacement $m_e \to 250\, m_e$ (no reduced mass) gives 118 eV.

3.3 Problem 3 — Natural width of the sodium D_2 line

In atomic physics "lifetime of a level" is the **mean life** τ. The number 16 ns is the standard mean life of Na $3p$. Use $\tau = 1.6 \times 10^{-8}$ s. Lower level is the ground state, so $\Gamma = 1/\tau$.

$$\Delta E = \frac{\hbar}{\tau} = 4.11 \times 10^{-8} \text{ eV}, \qquad \Delta\nu = \frac{1}{2\pi\tau} = 9.95 \text{ MHz}.$$

D_2 wavelength $\lambda = 589.0$ nm:

$$\Delta\lambda = \frac{\lambda^2}{c}\Delta\nu = 1.15 \times 10^{-4} \text{ Å}.$$

Wavenumber: $\Delta\bar{\nu} = 3.32 \times 10^{-4}$ cm^{-1}. Trap: using $\Delta\nu = 1/\tau = 62.5$ MHz (forgetting 2π). Using D_1 (589.6 nm) instead of D_2.

3.4 Problem 4 — Do the hydrogen series overlap?

Yes, but only in the infrared. Compare longest λ of series n with shortest λ (limit) of series $n + 1$.

Series	n_f	Longest λ	Limit λ_∞	Region
Lyman	1	121.6 nm	91.2 nm	UV
Balmer	2	656.3 nm	364.6 nm	vis / near UV
Paschen	3	1875 nm	820.4 nm	IR
Brackett	4	4051 nm	1458 nm	IR
Pfund	5	7460 nm	2279 nm	IR

- Lyman longest 121.6 $<$ Balmer shortest 364.6 $\to$ no overlap.
- Balmer longest 656.3 $<$ Paschen shortest 820.4 $\to$ no overlap.
- Paschen longest 1875 $>$ Brackett shortest 1458 $\to$ overlap.
- Brackett longest 4051 $>$ Pfund shortest 2279 $\to$ overlap.

Adjacent series overlap for $n_f \geq 3$.

3.5 Problem 5 — Two-laser pumping of H Rydberg states

Ground-state H. Laser 1 fixed at 11.5 eV. Laser 2 tunable. Populate a single $n \in \{20, 30, 40, 50\}$. Energy from $n = 1$ to n: $\Delta E_{1\to n} = 13.6(1 - 1/n^2)$ eV. Laser 1 already supplies 11.5 eV, so laser 2 supplies the rest. (11.5 eV itself is not resonant with a low-n Bohr level: $n \approx 2.55$.)

$$E_2 = 13.6\left(1 - \frac{1}{n^2}\right) - 11.5 \text{ eV}, \qquad \lambda_2 = \frac{1240 \text{ eV nm}}{E_2}.$$

Bohr radius: $r_n = n^2 a_0 = n^2 \times 0.529$ Å. Binding: $E_b = 13.6/n^2$ eV.

n	E_2 (eV)	λ_2 (nm)	r_n	E_b (eV)	$\lvert E_n - E_{n+1}\rvert$ (eV)
20	2.066	600.1	212 Å	0.03400	3.16×10^{-3}
30	2.085	594.7	476 Å	0.01511	9.59×10^{-4}
40	2.092	592.8	846 Å	0.00850	4.10×10^{-4}
50	2.095	591.9	1323 Å = 0.132 μm	0.00544	2.11×10^{-4}

Maximum laser linewidth so that only one n is populated: the tightest gap is $n = 50 \leftrightarrow 51$, $\Delta E_{50,51} = 2.11\times10^{-4}$ eV. The two photons add, so

$$\delta E_1 + \delta E_2 \; < \; \tfrac{1}{2}\Delta E_{50,51} = 1.06\times10^{-4}\ \text{eV} \approx 25.5\ \text{GHz}.$$

If split equally, each laser $\delta E \lesssim 5\times10^{-5}$ eV ($\approx$ 13 GHz). For laser 2 near 592 nm: $\delta\lambda_2 \approx 0.014$ nm = 0.14 Å.

3.6 Problem 6 — Wannier exciton in $\mathrm{Cu_2O}$

Electron + hole in a medium with $\varepsilon_r \approx 10$, $\mu \approx 0.7\,m_0$:

$$R_{\text{ex}} = 13.6\ \text{eV}\cdot\frac{\mu}{m_e}\cdot\frac{1}{\varepsilon_r^2} = 13.6\times0.7\times\frac{1}{100} = 95.2\ \text{meV}.$$

Levels measured from the band gap: $E_n = E_g - R_{\text{ex}}/n^2$.

(a) Binding energies R_{ex}/n^2: $n=2$: 23.8 meV; $n=3$: 10.6 meV; $n=4$: 5.95 meV; $n=5$: 3.81 meV.

(b) Absorption spectrum. Hydrogenic: a discrete series of lines at $h\nu = E_g - R_{\text{ex}}/n^2$, crowding toward E_g; oscillator strength falling $\sim 1/n^3$; a continuum for $h\nu > E_g$. The $n=1$ line is strongest and sits 95 meV below the gap. This is the yellow series of $\mathrm{Cu_2O}$ (Kittel).

3.7 Problem 7 — Scale spin–orbit from H $2p$ to Li^{2+} $5p$

Hydrogenic FS interval $\Delta E_{\text{fs}}(n,\ell,Z) \propto Z^4/[n^3\ell(\ell+1)]$. Both are p ($\ell=1$).

$$\frac{\Delta E(\mathrm{Li}^{2+},5p)}{\Delta E(\mathrm{H},2p)} = \left(\frac{3}{1}\right)^4\left(\frac{2}{5}\right)^3 = 81\times\frac{8}{125} = 5.184.$$

$$\Delta E = 0.3\times5.184 = 1.555\ \text{eV}.$$

Trap: using Z^3. Forgetting that Li^{2+} is hydrogenic with $Z=3$, not neutral Li.

3.8 Problem 8 — Doppler FWHM of H and D, $m = 5 \to n = 4$, $T = 5000$ K

Transition: Brackett-α,

$$\frac{1}{\lambda} = R\left(\frac{1}{16}-\frac{1}{25}\right) = R\cdot\frac{9}{400}.$$

$$\lambda_H = 40\,523\ \text{Å},\qquad \lambda_D = 40\,512\ \text{Å}.$$

FWHM:

$$\Delta\lambda_D = \frac{2\lambda_0}{c}\sqrt{\frac{2kT\ln2}{M}}.$$

$$\Delta\lambda_H = 2.04\ \text{Å},\qquad \Delta\lambda_D \approx \Delta\lambda_H/\sqrt{2} = 1.45\ \text{Å}.$$

(The $\lambda_D/\lambda_H \approx 1$ factor is a 0.03% correction; $\sqrt{M_H/M_D} = 1/\sqrt{2}$ dominates.)

3.9 Problem 9 — Na $3d \to 3p$: FS, anomalous Zeeman, Paschen–Back

Fine structure (zero field)

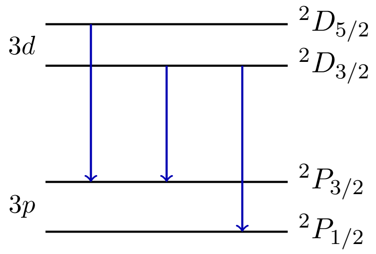
Valence $3d$ (^2D) $\to$ $3p$ (^2P).

$$3d:\ {}^2D_{5/2},\ {}^2D_{3/2},\qquad 3p:\ {}^2P_{3/2},\ {}^2P_{1/2}.$$

E1: $\Delta\ell = \pm 1$, $\Delta S = 0$, $\Delta J = 0, \pm 1$ not $0 \to 0$.

Upper → lower	ΔJ	Allowed?
${}^2D_{5/2} \to {}^2P_{3/2}$	-1	yes (strongest)
${}^2D_{5/2} \to {}^2P_{1/2}$	-2	**no**
${}^2D_{3/2} \to {}^2P_{3/2}$	0	yes
${}^2D_{3/2} \to {}^2P_{1/2}$	$+1$	yes

Three lines, not four. Regular FS (higher J higher in energy).



missing: ${}^2D_{5/2} \to {}^2P_{1/2}$ ($\Delta J = 2$)

Landé g-factors

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

Level	L	S	J	g_J
${}^2D_{5/2}$	2	1/2	5/2	$6/5 = 1.2$
${}^2D_{3/2}$	2	1/2	3/2	$4/5 = 0.8$
${}^2P_{3/2}$	1	1/2	3/2	$4/3 \approx 1.333$
${}^2P_{1/2}$	1	1/2	1/2	$2/3 \approx 0.667$

Weak-field energies: $E = E_{\text{FS}} + g_J \mu_B B\, m_J$.

Anomalous Zeeman

Take the strongest line ${}^2D_{5/2} \to {}^2P_{3/2}$. Upper: 6 sublevels, spacing $1.2\,\mu_B B$. Lower: 4 sublevels, spacing $1.333\,\mu_B B$. Selection $\Delta m_J = 0, \pm 1$:

- π ($\Delta m_J = 0$): vertical arrows.
- $\sigma_\pm$ ($\Delta m_J = \pm 1$): arrows one step right/left.

Viewed $\perp$ $\mathbf{B}$: π linear $\parallel$ $\mathbf{B}$, σ linear $\perp$ $\mathbf{B}$. Viewed along $\mathbf{B}$: only σ, circular; π is absent.

Repeat the same picture for the other two FS lines.

Paschen–Back ($\mu_B B \gg E_{\text{FS}}$)

Decouple: good quantum numbers m_ℓ, m_s with $E = \mu_B B(m_\ell + 2m_s)$. Optical: $\Delta m_\ell = 0, \pm 1$, $\Delta m_s = 0$ (light does not flip spin). Then

$$\Delta E = \mu_B B\, \Delta m_\ell \in \{-\mu_B B,\ 0,\ +\mu_B B\}.$$

Normal Lorentz triplet. Polarisation: π ($\Delta m_\ell = 0$), $\sigma_\pm$ ($\Delta m_\ell = \pm 1$). Residual SO merely weakly splits each of the three.

3.10 Problem 10 — LS terms for $\ell_1 = 1$, $\ell_2 = 2$ (nonequivalent)

Nonequivalent, so Pauli does not kill terms. $L = |\ell_1 - \ell_2| \ldots \ell_1 + \ell_2 = 1, 2, 3 \Rightarrow P, D, F$. $S = 0, 1 \Rightarrow$ singlets and triplets.

Term	L	S	J	Symbols
singlet P	1	0	1	${}^1P_1$
singlet D	2	0	2	${}^1D_2$
singlet F	3	0	3	${}^1F_3$
triplet P	1	1	0,1,2	${}^3P_0, {}^3P_1, {}^3P_2$
triplet D	2	1	1,2,3	${}^3D_1, {}^3D_2, {}^3D_3$
triplet F	3	1	2,3,4	${}^3F_2, {}^3F_3, {}^3F_4$

3.11 Problem 11 — Stern–Gerlach, potassium, P_2P_3

K ground state $4s\ {}^2S_{1/2}$: two beams, $\mu_z = \pm\mu_B$. $T = 700$ K, $\partial B_z/\partial z = 10^3\ \mathrm{T\,m^{-1}}$, $L = 0.1$ m, $\ell = 1$ m.

Geometry. Magnet of length L along the beam, then a field-free flight of length ℓ to the plate. Spots P_2, P_3 are the two components. (If your figure swaps the labels, swap $L \leftrightarrow \ell$ in the last line.)

Force $F_z = \mu_B \partial B_z/\partial z$, acceleration $a = F_z/M$. Time in the magnet $t_1 = L/v$. Deflection at magnet exit $\frac{1}{2}at_1^2$, plus extra $at_1 \cdot (\ell/v)$ during the drift:

$$s = \frac{\mu_B}{M}\frac{\partial B_z}{\partial z}\frac{L}{v^2}\left(\ell + \frac{L}{2}\right).$$

Most probable thermal speed $v = \sqrt{2kT/M}$. Potassium: $M = 39.1\,\mathrm{u} = 6.49 \times 10^{-26}$ kg.

$$v = 546\ \mathrm{m\,s^{-1}}, \qquad a = 1.43 \times 10^5\ \mathrm{m\,s^{-2}}.$$

$$s = 0.0504\ \mathrm{m}, \qquad P_2P_3 = 2s = 10.1\ \mathrm{cm}.$$

If the figure takes $\ell = 1$ m as the magnet and $L = 0.1$ m as the drift: $2s \approx 58$ cm. Quote the geometry you drew. The physically typical SG deflection with a 10 cm magnet is the 10 cm answer.

Traps: using $J = 3/2$ (excited K) $\to$ 4 spots. Using $g = 1$ instead of $g = 2$ for ${}^2S_{1/2}$. Trying to do SG on electrons.

3.12 Problem 12 — Linear Stark of H, $n = 3$, and $n = 3 \to 2$

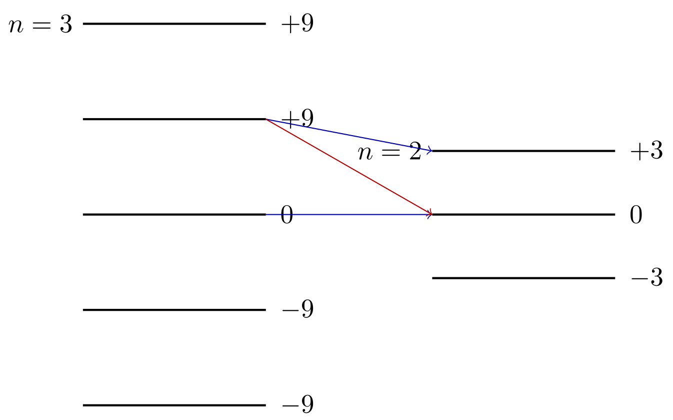
Parabolic formula

$$\Delta E = \frac{3}{2}n(n_1 - n_2)\, ea_0\, \mathcal{E}, \qquad n = n_1 + n_2 + |m| + 1.$$

$n = 3$ (9 orbital states $\to$ 5 energies)

n_1	n_2	m	$n_1 - n_2$	$\Delta E/(ea_0\mathcal{E})$	degeneracy
2	0	0	+2	+9	1
1	0	±1	+1	+9/2	2
1	1	0	0	0	1
0	0	±2	0	0	2
0	1	±1	−1	−9/2	2
0	2	0	−2	−9	1

Five levels: ± 9, $\pm 9/2$, 0 (units $ea_0\mathcal{E}$). Degeneracies 1, 2, 3, 2, 1. $n = 2$: three levels, $\Delta E = 0, \pm 3\, ea_0\mathcal{E}$.



π: $\Delta m = 0$; σ: $\Delta m = \pm 1$

E1: $\Delta m = 0$ (π, $\mathbf{E}_{\text{light}} \parallel \boldsymbol{\mathcal{E}}$) and $\Delta m = \pm 1$ (σ). The pattern is symmetric about the unshifted line (linear Stark). Draw the five $n = 3$ and three $n = 2$ bars, π arrows in the same-m column, σ arrows with $\Delta m = \pm 1$.

3.13 Problem 13 — Pt ground term, μ, weak-field span

Configuration: [Xe] $4f^{14}5d^9 6s^1$. Closed $4f^{14}$ is inert. Treat $5d^9$ as a d-**hole** ($L = 2, S = 1/2$) plus $6s^1$ ($l = 0, s = 1/2$):

$$L = 2, \qquad S = 0 \text{ or } 1 \quad \Rightarrow \quad {}^1D \text{ and } {}^3D.$$

Hund: max S first $\Rightarrow$ triplet ${}^3D$ is lower. The d^9 shell is more than half full $\Rightarrow$ inverted FS: highest J lies lowest.

$$J = 3, 2, 1 \quad \Rightarrow \quad \text{GS } = {}^3D_3.$$

$$g_J = 1 + \frac{12 + 2 - 6}{2 \cdot 12} = 1 + \frac{8}{24} = \frac{4}{3}.$$

$$\mu = g_J \sqrt{J(J+1)}\, \mu_B = \frac{4}{3}\sqrt{12}\, \mu_B = \frac{8\sqrt{3}}{3}\, \mu_B \approx 4.619\, \mu_B.$$

Weak field: $E = g_J \mu_B B m_J$, $m_J = -3, \ldots, +3$. Extreme components $\Delta m_J = 6$:

$$\Delta E_{\text{span}} = 6 \cdot \frac{4}{3} \cdot \mu_B B = 8\, \mu_B B.$$

3.14 Problem 14 — jj coupling, $(5/2)^4$

Four equivalent electrons in a $j = 5/2$ subshell. The subshell holds $2j + 1 = 6$ electrons, so $(5/2)^4$ is two **holes** in a closed $j = 5/2$ shell. Two equivalent $j = 5/2$ fermions: Pauli $\Rightarrow$ only even J,

$$J = 0, 2, 4.$$

Hole–particle symmetry: $(5/2)^4$ has the same list as $(5/2)^2$. NIST: $(5/2)^2$ and $(5/2)^4$: $J = 0, 2, 4$.

3.15 Problem 15 — Helium $|{\downarrow\downarrow}\rangle$ is an eigenstate of S^2 and S_z

Two-electron spin-down product

$$\chi = |\beta(1)\beta(2)\rangle.$$

$\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2$, $\mathbf{S}_i = (\hbar/2)\boldsymbol{\sigma}_i$.

S_z. $\sigma_z \beta = -\beta$, so

$$S_z \chi = \frac{\hbar}{2}(\sigma_{z1} + \sigma_{z2})\chi = -\hbar\, \chi.$$

Eigenvalue $-\hbar \Rightarrow M_S = -1$.

S^2**.** Use $S^2 = S_1^2 + S_2^2 + 2\mathbf{S}_1\cdot\mathbf{S}_2$ with $S_i^2 = (3/4)\hbar^2$ and

$$\mathbf{S}_1\cdot\mathbf{S}_2 = \frac{\hbar^2}{4}\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2.$$

Act on $\chi = |{\downarrow\downarrow}\rangle$:

$$\begin{aligned} \sigma_x|{\downarrow}\rangle &= |{\uparrow}\rangle &&\Rightarrow\ \sigma_{x1}\sigma_{x2}\chi = |{\uparrow\uparrow}\rangle,\\ \sigma_y|{\downarrow}\rangle &= -i|{\uparrow}\rangle &&\Rightarrow\ \sigma_{y1}\sigma_{y2}\chi = -|{\uparrow\uparrow}\rangle,\\ \sigma_z|{\downarrow}\rangle &= -|{\downarrow}\rangle &&\Rightarrow\ \sigma_{z1}\sigma_{z2}\chi = +\chi. \end{aligned}$$

$$\boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2\,\chi = |{\uparrow\uparrow}\rangle - |{\uparrow\uparrow}\rangle + \chi = \chi.$$

Hence $\mathbf{S}_1\cdot\mathbf{S}_2\,\chi = (\hbar^2/4)\chi$, and

$$S^2\chi = \left(\tfrac{3}{4}+\tfrac{3}{4}+\tfrac{1}{2}\right)\hbar^2\chi = 2\hbar^2\chi = 1(1+1)\hbar^2\chi.$$

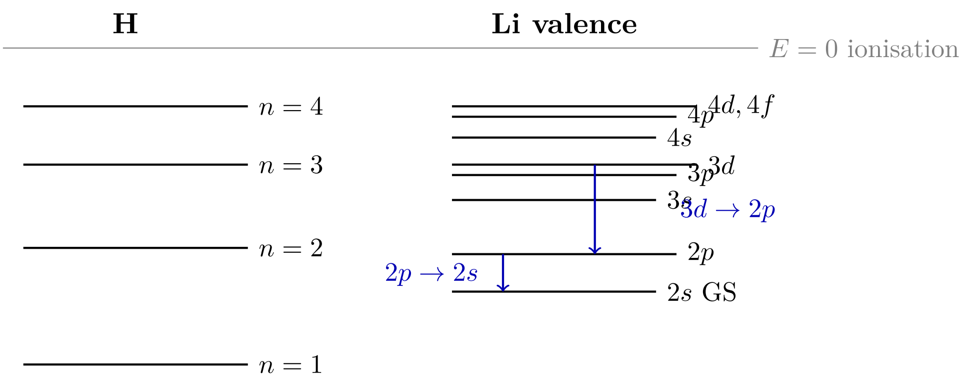
So χ is $|S=1, M_S=-1\rangle$, a triplet function, simultaneous eigenstate of S^2 and S_z.

3.16 Problem 16 — Li valence levels vs H

Quantum defects: $\delta_s = 0.4$, $\delta_p = 0.04$, $\delta_d = \delta_f = 0$.

$$E_{n\ell}(\text{Li}) = -13.6\ \text{eV}/(n-\delta_\ell)^2, \qquad E_n(\text{H}) = -13.6\ \text{eV}/n^2.$$

Effective quantum numbers for Li: $n_s^* = n - 0.4$, $n_p^* = n - 0.04$, $n_{d,f}^* = n$.



Forbidden: $3s \to 2s$ ($\Delta\ell = 0$), $3d \to 2s$ ($\Delta\ell = 2$). Li $1s^2$ is a filled core, not the valence.

3.17 Problem 17 — Unsold's theorem, prove for $\ell = 1$

Statement. For a filled ℓ-subshell (or the sum over m at fixed ℓ),

$$\sum_{m=-\ell}^{\ell} |Y_{\ell m}(\theta,\phi)|^2 = \frac{2\ell+1}{4\pi},$$

independent of angle. The charge density of a closed subshell is spherically symmetric.

Proof, $\ell = 1$, using the given Φ, Θ.

$$\Phi = \frac{1}{\sqrt{2\pi}}, \qquad \Theta_{10} = \frac{\sqrt{6}}{2}\cos\theta, \qquad \Theta_{1\pm1} = \frac{\sqrt{3}}{2}\sin\theta.$$

$$|Y_{10}|^2 = \frac{3}{4\pi}\cos^2\theta, \qquad |Y_{1\pm1}|^2 = \frac{3}{8\pi}\sin^2\theta.$$

$$\sum_{m=-1}^{1}|Y_{1m}|^2 = \frac{3}{4\pi}\cos^2\theta + 2\cdot\frac{3}{8\pi}\sin^2\theta = \frac{3}{4\pi}(\cos^2\theta + \sin^2\theta) = \frac{3}{4\pi} = \frac{2\cdot1+1}{4\pi}.$$

Independent of θ and ϕ.

3.18 Problem 18 — True / False

All seven statements are FALSE.

1. At small r, for $\ell \neq 0$, $R_{nl} \propto r^{\ell+1}$. **FALSE.** $R_{nl} \propto r^{\ell}$. It is $u(r) = rR_{nl}$ that goes as $r^{\ell+1}$.
2. Series limit of Lyman of H lies in the visible. **FALSE.** Lyman limit is 91.2 nm, vacuum UV.
3. Stimulated-emission rates are independent of the intensity of the incident radiation. **FALSE.** Stimulated rate $= B_{21}\rho(\nu)$. Spontaneous A_{21} is the intensity-independent one.
4. In alkali atoms ℓ-degeneracy is removed because the effective potential is not spherically symmetric. **FALSE.** The central-field potential **is** spherically symmetric (that is why m remains degenerate). ℓ-degeneracy dies because V_{eff} is not Coulomb $1/r$.
5. The energy of an Auger electron depends on the energy of the radiation which created the initial vacancy. **FALSE.** $K_{\text{Auger}} = E_{\text{hole}} - E_{\text{filler}} - E_{\text{ejected}}$. Photoelectrons *do* remember the photon; Auger electrons do not.
6. Atomic transitions forbidden under the dipole approximation are 'strictly forbidden'. **FALSE.** They proceed by M1, E2, two-photon decay, collisions, . . . Strictly forbidden is reserved for exact-symmetry bans (e.g. $J = 0 \to 0$).
7. Electrons in high Rydberg states are very strongly bound. **FALSE.** $E_b = 13.6\ \text{eV}/n^2 \to 0$. They are weakly bound.

The two people miss under exam stress are (i) (confusing R with u) and (iv) (confusing spherical with Coulomb).

4 Last-hour drill

Read this once in the morning. Do not open a textbook.

60-second constants

$$R_\infty = 109737 \text{ cm}^{-1}, \quad R_H = 109678 \text{ cm}^{-1}, \quad 13.6 \text{ eV}, \quad a_0 = 0.529 \text{ Å}$$

$$R_M = R_\infty/(1 + m_e/M), \quad m_p/m_e = 1836, \quad M_D/m_e = 3670$$

Na $D_2 = 589.0$ nm $= 3\,^2P_{3/2} \to 3\,^2S_{1/2}$. $D_1 = 589.6$ nm.

Diagrams you must be able to draw blind

1. H series. Lyman UV, Balmer vis, rest IR. Overlap starts at Paschen/Brackett.
2. Li vs H. ns pulled down ($\delta = 0.4$), np a little ($\delta = 0.04$), nd, nf on H. Arrows: $2p \to 2s$, $3d \to 2p$.
3. Na $3d \to 3p$ FS. Three lines. Missing $^2D_{5/2} \to {}^2P_{1/2}$.
4. Anomalous Zeeman of a doublet (Na D): $^2P_{3/2}$ (4), $^2P_{1/2}$ (2), $^2S_{1/2}$ (2). π $\Delta m = 0$, σ $\Delta m = \pm 1$.
5. Paschen–Back. $E = \mu_B B(m_L + 2m_S)$. Triplet. $\Delta m_S = 0$.
6. Stark $n = 3$. Five bars at $0, \pm 4.5, \pm 9$ ($ea_0\mathcal{E}$). $n = 2$: three bars at $0, \pm 3$.
7. SG. Two spots for $^2S_{1/2}$. $F_z = \mu_z \partial B_z/\partial z$. Not for free e^-.

g_J flash

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}$$

$^2S_{1/2} \to 2$, $^2P_{1/2} \to 2/3$, $^2P_{3/2} \to 4/3$, $^2D_{3/2} \to 4/5$, $^2D_{5/2} \to 6/5$, singlets $\to 1$.

Widths

Natural (mean life τ, lower level stable): $\Delta\nu = 1/(2\pi\tau)$. Na 16 ns $\Rightarrow$ 10 MHz.
Doppler FWHM: $\Delta\lambda = (2\lambda/c)\sqrt{2kT\ln 2/M}$. Scales $\sqrt{T/M}\,\lambda$.
Stimulated $\propto I$. Spontaneous does not.

Scaling laws

Quantity	Scale
Bohr energy	$\mu Z^2/n^2$
Bohr radius	$n^2/\mu Z$
Fine-structure interval	$Z^4/[n^3\ell(\ell+1)]$
Linear Stark	$n(n_1 - n_2)ea_0\mathcal{E}$
Weak Zeeman	$g_J \mu_B B m_J$
Rydberg spacing	27.2 eV$/n^3$
Exciton Rydberg	$13.6\,(\mu/m_e)/\varepsilon_r^2$

Five exam traps (the difference between 90 and 100)

1. **Reduced mass.** Any mention of D, μ, π, or R_∞ vs R_H is a reduced-mass question.
2. Z^4 for fine structure, Z^2 for Bohr energies, $1/n^3$ for Rydberg spacings and SO.
3. $\Delta J = 2$ is E1-forbidden. Count lines before drawing them.
4. Geometry of SG. Write $s = \frac{\mu}{M}(\partial B/\partial z)(L/v^2)(\ell + L/2)$ and $2s$. State which length is the magnet.
5. Spherical $\neq$ Coulomb. m-degeneracy from spherical; ℓ-degeneracy from $1/r$.

If a derivation is asked

Rydberg from correspondence. Classical $E(\omega) + \omega = 4\pi Rc/n^3 \Rightarrow R_\infty = me^4/(8\varepsilon_0^2 h^3 c)$.
SG force. $U = -\mu \cdot B \Rightarrow F_z = \mu_z \partial B_z/\partial z$. Drop F_x, F_y.
SO. $\mathbf{B}_L \propto \mathbf{L}/r^3$, Thomas 1/2, $\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(J^2 - L^2 - S^2)$, $\langle r^{-3} \rangle \propto Z^3/[n^3\ell(\ell + 1/2)(\ell + 1)]$.
Dipole Δm. $z \Rightarrow \int e^{i(m_a - m_b)\phi} d\phi \Rightarrow \Delta m = 0$; $x \pm iy \Rightarrow \Delta m = \pm 1$. Parity of $Y_{\ell m}$ is $(-1)^\ell$.

That is the whole quiz. Draw the diagrams once more, then sleep.